

LEARNING MODULE

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Security Market Indexes

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LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	describe a security market index
☐	calculate and interpret the value, price return, and total return of an index
☐	describe the choices and issues in index construction and management
☐	compare the different weighting methods used in index construction
☐	calculate and analyze the value and return of an index given its weighting method
☐	describe rebalancing and reconstitution of an index
☐	describe uses of security market indexes
☐	describe types of equity indexes
☐	compare types of security market indexes
☐	describe types of fixed-income indexes
☐	describe indexes representing alternative investments

INTRODUCTION

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Investors gather and analyze vast amounts of information about security markets on a continual basis. Because this work can be both time consuming and data intensive, investors often use a single measure that consolidates this information and reflects the performance of an entire security market.

Security market indexes were first introduced as a simple measure to reflect the performance of the US stock market. Since then, security market indexes have evolved into important multi-purpose tools that help investors track the performance of various security markets, estimate risk, and evaluate the performance of investment managers. They also form the basis for new investment products.

in·dex, *noun* (*pl.in-dex-es or in-di-ces*) Latin *indic-*, *index*, from *indicare* to indicate: an indicator, sign, or measure of something.

ORIGIN OF MARKET INDEXES

Investors had access to regularly published data on individual security prices in London as early as 1698, but nearly 200 years passed before they had access to a simple indicator to reflect security market information. To give readers a sense of how the US stock market in general performed on a given day, publishers Charles H. Dow and Edward D. Jones introduced the Dow Jones Average, the world's first security market index, in 1884. The index, which appeared in *The Customers' Afternoon Letter*, consisted of the stocks of nine railroads and two industrial companies. It eventually became the Dow Jones Transportation Average. Convinced that industrial companies, rather than railroads, would be "the great speculative market" of the future, Dow and Jones introduced a second index in May 1896—the Dow Jones Industrial Average (DJIA). It had an initial value of 40.94 and consisted of 12 stocks from major US industries. Today, investors can choose from among thousands of indexes to measure and monitor different security markets and asset classes.

This reading is organized as follows. Section 2 defines a security market index and explains how to calculate the price return and total return of an index for a single period and over multiple periods. Section 3 describes how indexes are constructed and managed. Section 4 discusses the use of market indexes. Sections 5, 6, and 7 discuss various types of indexes, and the final section summarizes the reading. Practice problems follow the conclusions and summary.

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INDEX DEFINITION AND CALCULATIONS OF VALUE AND RETURNS

- ☐ describe a security market index
- ☐ calculate and interpret the value, price return, and total return of an index

A **security market index** represents a given security market, market segment, or asset class. Most indexes are constructed as portfolios of marketable securities.

The value of an index is calculated on a regular basis using either the actual or estimated market prices of the individual securities, known as **constituent securities**, within the index. For each security market index, investors may encounter two versions of the same index (i.e., an index with identical constituent securities and weights): one version based on price return and one version based on total return. As the name suggests, a **price return index**, also known as a **price index**, reflects *only* the prices of the constituent securities within the index. A **total return index**, in contrast, reflects not only the prices of the constituent securities but also the reinvestment of all income received since inception.

At inception, the values of the price and total return versions of an index are equal. As time passes, however, the value of the total return index, which includes the reinvestment of all dividends and/or interest received, will exceed the value of the price return index by an increasing amount. A look at how the values of each version are calculated over multiple periods illustrates why.

The value of a price return index is calculated as:

$$V_{PRI} = \frac{\sum_{i=1}^N n_i P_i}{D} \quad (1)$$

where

V_{PRI} = the value of the price return index

n_i = the number of units of constituent security i held in the index portfolio

N = the number of constituent securities in the index

P_i = the unit price of constituent security i

D = the value of the divisor

The **divisor** is a number initially chosen at inception. It is frequently chosen so that the price index has a convenient initial value, such as 1,000. The index provider then adjusts the value of the divisor as necessary to avoid changes in the index value that are unrelated to changes in the prices of its constituent securities. For example, when changing index constituents, the index provider may adjust the divisor so that the value of the index with the new constituents equals the value of the index prior to the changes.

Index return calculations, like calculations of investment portfolio returns, may measure price return or total return. **Price return** measures only price appreciation or percentage change in price. **Total return** measures price appreciation plus interest, dividends, and other distributions.

Calculation of Single-Period Returns

For a security market index, price return can be calculated in two ways: either the percentage change in value of the price return index, or the weighted average of price returns of the constituent securities. The price return of an index can be expressed as:

$$PR_I = \frac{V_{PRI1} - V_{PRI0}}{V_{PRI0}} \quad (2)$$

where

PR_I = the price return of the index portfolio (as a decimal number, i.e., 12 percent is 0.12)

V_{PRI1} = the value of the price return index at the end of the period

V_{PRI0} = the value of the price return index at the beginning of the period

Similarly, the price return of each constituent security can be expressed as:

$$PR_i = \frac{P_{i1} - P_{i0}}{P_{i0}} \quad (3)$$

where

PR_i = the price return of constituent security i (as a decimal number)

P_{i1} = the price of constituent security i at the end of the period

P_{i0} = the price of constituent security i at the beginning of the period

Because the price return of the index equals the weighted average of price returns of the individual securities, we can write:

$$PR_I = \sum_{i=1}^N w_i PR_i = \sum_{i=1}^N w_i \left(\frac{P_{i1} - P_{i0}}{P_{i0}} \right) \quad (4)$$

where:

PR_I = the price return of index portfolio (as a decimal number)

PR_i = the price return of constituent security i (as a decimal number)

N = the number of individual securities in the index

w_i = the weight of security i (the fraction of the index portfolio allocated to security i)

P_{i1} = the price of constituent security i at the end of the period

P_{i0} = the price of constituent security i at the beginning of the period

Equation 4 can be rewritten simply as:

$$PR_I = w_1 PR_1 + w_2 PR_2 + \dots + w_N PR_N \quad (5)$$

where

PR_I = the price return of index portfolio (as a decimal number)

PR_i = the price return of constituent security i (as a decimal number)

w_i = the weight of security i (the fraction of the index portfolio allocated to security i)

N = the number of securities in the index

Total return measures price appreciation plus interest, dividends, and other distributions. Thus, the **total return** of an index is the price appreciation, or change in the value of the price return index, plus income (dividends and/or interest) over the period, expressed as a percentage of the beginning value of the price return index. The total return of an index can be expressed as:

$$TR_I = \frac{V_{PRI1} - V_{PRI0} + Inc_I}{V_{PRI0}} \quad (6)$$

where

TR_I = the total return of the index portfolio (as a decimal number)

V_{PRI1} = the value of the price return index at the end of the period

V_{PRI0} = the value of the price return index at the beginning of the period

Inc_I = the total income (dividends and/or interest) from all securities in the index held over the period

The total return of an index can also be calculated as the weighted average of total returns of the constituent securities. The total return of each constituent security in the index is calculated as:

$$TR_i = \frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}} \quad (7)$$

where

TR_i = the total return of constituent security i (as a decimal number)

P_{1i} = the price of constituent security i at the end of the period

P_{0i} = the price of constituent security i at the beginning of the period

Inc_i = the total income (dividends and/or interest) from security i over the period

Because the total return of an index can be calculated as the weighted average of total returns of the constituent securities, we can express total return as:

$$TR_I = \sum_{i=1}^N w_i TR_i = \sum_{i=1}^N w_i \left(\frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}} \right) \quad (8)$$

Equation 8 can be rewritten simply as

$$TR_I = w_1 TR_1 + w_2 TR_2 + \dots + w_N TR_N \quad (9)$$

where

TR_I = the total return of the index portfolio (as a decimal number)

TR_i = the total return of constituent security i (as a decimal number)

w_i = the weight of security i (the fraction of the index portfolio allocated to security i)

N = the number of securities in the index

Calculation of Index Values over Multiple Time Periods

The calculation of index values over multiple time periods requires geometrically linking the series of index returns. With a series of price returns for an index, we can calculate the value of the price return index with the following equation:

$$V_{PRIT} = V_{PRI0}(1 + PR_{I1})(1 + PR_{I2})\dots(1 + PR_{IT}) \quad (10)$$

where

V_{PRI0} = the value of the price return index at inception

V_{PRIT} = the value of the price return index at time t

PR_{IT} = the price return (as a decimal number) on the index over period t , $t = 1, 2, \dots, T$

For an index with an inception value set to 1,000 and price returns of 5 percent and 3 percent for Periods 1 and 2 respectively, the values of the price return index would be calculated as follows:

Period	Return (%)	Calculation	Ending Value
0		1,000(1.00)	1,000.00
1	5.00	1,000(1.05)	1,050.00
2	3.00	1,000(1.05)(1.03)	1,081.50

Similarly, the series of total returns for an index is used to calculate the value of the total return index with the following equation:

$$V_{TRIT} = V_{TRIO}(1 + TR_{I1})(1 + TR_{I2}) \dots (1 + TR_{IT}) \quad (11)$$

where

V_{TRIO} = the value of the index at inception

V_{TRIT} = the value of the total return index at time t

TR_{IT} = the total return (as a decimal number) on the index over period t , $t = 1, 2, \dots, T$

Suppose that the same index yields an additional 1.5 percent return from income in Period 1 and an additional 2.0 percent return from income in Period 2, bringing the total returns for Periods 1 and 2, respectively, to 6.5 percent and 5 percent. The values of the total return index would be calculated as follows:

Period	Return (%)	Calculation	Ending Value
0		1,000(1.00)	1,000.00
1	6.50	1,000(1.065)	1,065.00
2	5.00	1,000(1.065)(1.05)	1,118.25

As illustrated above, as time passes, the value of the total return index, which includes the reinvestment of all dividends and/or interest received, exceeds the value of the price return index by an increasing amount.

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INDEX CONSTRUCTION

- ☐ describe the choices and issues in index construction and management
- ☐ compare the different weighting methods used in index construction
- ☐ calculate and analyze the value and return of an index given its weighting method

Constructing and managing a security market index is similar to constructing and managing a portfolio of securities. Index providers must decide the following:

1. Which target market should the index represent?
2. Which securities should be selected from that target market?
3. How much weight should be allocated to each security in the index?
4. When should the index be rebalanced?
5. When should the security selection and weighting decision be re-examined?

Target Market and Security Selection

The first decision in index construction is identifying the target market, market segment, or asset class that the index is intended to represent. The target market may be defined very broadly or narrowly. It may be based on asset class (e.g., equities, fixed income, real estate, commodities, hedge funds); geographic region (e.g., Japan, South Africa, Latin America, Europe); the exchange on which the securities are traded (e.g., Shanghai, Toronto, Tokyo), and/or other characteristics (e.g., economic sector, company size, investment style, duration, or credit quality).

The target market determines the investment universe and the securities available for inclusion in the index. Once the investment universe is identified, the number of securities and the specific securities to include in the index must be determined. The constituent securities could be nearly all those in the target market or a representative sample of the target market. Some equity indexes, such as the S&P 500 Index and the FTSE 100, fix the number of securities included in the index and indicate this number in the name of the index. Other indexes allow the number of securities to vary to reflect changes in the target market or to maintain a certain percentage of the target market. For example, the Tokyo Stock Price Index (TOPIX) represents and includes all of the largest stocks, known as the First Section, listed on the Tokyo Stock Exchange. To be included in the First Section—and thus the TOPIX—stocks must meet certain criteria, such as the number of shares outstanding, the number of shareholders, and market capitalization. Stocks that no longer meet the criteria are removed from the First Section and also the TOPIX. Objective or mechanical rules determine the constituent securities of most, but not all, indexes. The S&P Bombay Stock Exchange Sensitive Index, also called the S&P BSE SENSEX and the S&P 500, for example, use a selection committee and more subjective decision-making rules to determine constituent securities.

Index Weighting

The weighting decision determines how much of each security to include in the index and has a substantial impact on an index's value. Index providers use a number of methods to weight the constituent securities in an index. Indexes can be price weighted, equal weighted, market-capitalization weighted, or fundamentally weighted. Each weighting method has its advantages and disadvantages.

Price Weighting

The simplest method to weight an index and the one used by Charles Dow to construct the Dow Jones Industrial Average is **price weighting**. In price weighting, the weight on each constituent security is determined by dividing its price by the sum of all the prices of the constituent securities. The weight is calculated using the following formula:

$$w_i^P = \frac{P_i}{\sum_{i=1}^N P_i} \quad (12)$$

Exhibit 1 illustrates the values, weights, and single-period returns following inception of a price-weighted equity index with five constituent securities. The value of the price-weighted index is determined by dividing the sum of the security values (101.50) by the divisor, which is typically set at inception to equal the initial number of securities in the index. Thus, in our example, the divisor is 5 and the initial value of the index is calculated as $101.50 \div 5 = 20.30$.

Exhibit 1: Example of a Price-Weighted Equity Index

Security	Shares in Index	Value		BOP Weight (%)	Value		Dividends Per Share	Dividends	EOP Price	Price Return (%)	Total Return (%)	BOP Weight × Price Return (%)	BOP Weight × Total Return (%)	EOP Weight (%)
		(Shares × BOP Price)	(Shares × EOP Price)		(Shares × EOP Price)	Total Dividends								
A	1	50.00	50.00	49.26	55.00	0.75	55.00	0.75	55.00	10.00	11.50	4.93	5.66	52.38
B	1	25.00	25.00	24.63	22.00	0.10	22.00	0.10	22.00	-12.00	-11.60	-2.96	-2.86	20.95
C	1	12.50	12.50	12.32	8.00	0.00	8.00	0.00	8.00	-36.00	-36.00	-4.43	-4.43	7.62
D	1	10.00	10.00	9.85	14.00	0.05	14.00	0.05	14.00	40.00	40.50	3.94	3.99	13.33
E	1	4.00	4.00	3.94	6.00	0.00	6.00	0.00	6.00	50.00	50.00	1.97	1.97	5.72
Total		101.50	101.50	100.00					105.00			3.45	4.33	100.00
Index Value		20.30							21.00	3.45	4.33			

Divisor = 5

BOP = Beginning of period

EOP = End of period

Type of Index	BOP Value	Return (%)	EOP Value
Price Return	20.30	3.45	21.00
Total Return	20.30	4.33	21.18

As illustrated in this exhibit, Security A, which has the highest price, also has the highest weighting and thus will have the greatest impact on the return of the index. Note how both the price return and the total return of the index are calculated on the basis of the corresponding returns on the constituent securities.

A property unique to price-weighted indexes is that a stock split on one constituent security changes the weights on all the securities in the index.¹ To prevent the stock split and the resulting new weights from changing the value of the index, the index provider must adjust the value of the divisor as illustrated in Exhibit 2. Given a 2-for-1 split in Security A, the divisor is adjusted by dividing the sum of the constituent prices *after* the split (77.50) by the value of the index *before* the split (21.00). This adjustment results in changing the divisor from 5 to 3.69 so that the index value is maintained at 21.00.

The primary advantage of price weighting is its simplicity. The main disadvantage of price weighting is that it results in arbitrary weights for each security. In particular, a stock split in any one security causes arbitrary changes in the weights of all the constituents' securities.

Exhibit 2: Impact of 2-for-1 Split in Security A

Security	Price before Split	Weight before Split (%)	Price after Split	Weight after Split (%)
A	55.00	52.38	27.50	35.48
B	22.00	20.95	22.00	28.39
C	8.00	7.62	8.00	10.32
D	14.00	13.33	14.00	18.07
E	6.00	5.72	6.00	7.74
Total	105.00	100.00	77.50	100.00
Divisor	5.00		3.69	
Index Value	21.00		21.00	

Equal Weighting

Another simple index weighting method is **equal weighting**. This method assigns an equal weight to each constituent security at inception. The weights are calculated as:

$$w_i^E = \frac{1}{N} \quad (13)$$

where

w_i = fraction of the portfolio that is allocated to security i or weight of security i

N = number of securities in the index

To construct an equal-weighted index from the five securities in Exhibit 1, the index provider allocates one-fifth (20 percent) of the value of the index (at the beginning of the period) to each security. Dividing the value allocated to each security by each security's individual share price determines the number of shares of each security to include in the index. Unlike a price-weighted index, where the weights are arbitrarily determined by the market prices, the weights in an equal-weighted index are arbitrarily assigned by the index provider.

¹ A stock split is an increase in the number of shares outstanding and a proportionate decrease in the price per share such that the total market value of equity, as well as investors' proportionate ownership in the company, does not change.

Exhibit 3 illustrates the values, weights, and single-period returns following inception of an equal-weighted index with the same constituent securities as those in Exhibit 1. This example assumes a beginning index portfolio value of 10,000 (i.e., an investment of 2,000 in each security). To set the initial value of the index to 1,000, the divisor is set to 10 ($10,000 \div 10 = 1,000$).

Exhibit 1 and Exhibit 3 demonstrate how different weighting methods result in different returns. The 10.4 percent price return of the equal-weighted index shown in Exhibit 3 differs significantly from the 3.45 percent price return of the price-weighted index in Exhibit 1.

Like price weighting, the primary advantage of equal weighting is its simplicity. Equal weighting, however, has a number of disadvantages. First, securities that constitute the largest fraction of the target market value are underrepresented, and securities that constitute a small fraction of the target market value are overrepresented. Second, after the index is constructed and the prices of constituent securities change, the index is no longer equally weighted. Therefore, maintaining equal weights requires frequent adjustments (rebalancing) to the index.

Market-Capitalization Weighting

In **market-capitalization weighting**, or value weighting, the weight on each constituent security is determined by dividing its market capitalization by the total market capitalization (the sum of the market capitalization) of all the securities in the index. Market capitalization or value is calculated by multiplying the number of shares outstanding by the market price per share.

The market-capitalization weight of security i is:

$$w_i^M = \frac{Q_i P_i}{\sum_{j=1}^N Q_j P_j} \quad (14)$$

where

w_i = fraction of the portfolio that is allocated to security i or weight of security i

Q_i = number of shares outstanding of security i

P_i = share price of security i

N = number of securities in the index

Exhibit 4 illustrates the values, weights, and single-period returns following inception of a market-capitalization-weighted index for the same five-security market. Security A, with 3,000 shares outstanding and a price of 50 per share, has a market capitalization of 150,000 or 26.29 percent ($150,000/570,500$) of the entire index portfolio. The resulting index weights in the exhibit reflect the relative value of each security as measured by its market capitalization.

As shown in Exhibit 1, Exhibit 3, and Exhibit 4, the weighting method affects the index's returns. The price and total returns of the market-capitalization index in Exhibit 4 (1.49 percent and 2.13 percent, respectively) differ significantly from those of the price-weighted (3.45 percent and 4.33 percent, respectively) and equal-weighted (10.40 percent and 10.88 percent respectively) indexes. To understand the source and magnitude of the difference, compare the weights and returns of each security under each of the weighting methods. The weight of Security A, for example, ranges from 49.26 percent in the price-weighted index to 20 percent in the equal-weighted index. With a price return of 10 percent, Security A contributes 4.93 percent to the price return of the price-weighted index, 2.00 percent to the price return of the equal-weighted index, and 2.63 percent to the price return of the market-capitalization-weighted index. With

Exhibit 4: Example of a Market-Capitalization-Weighted Equity Index

Stock	Shares Out-standing	BOP Market Cap	BOP Weight (%)	EOP Price	Dividends Per Share	EOP Market Cap	Total Dividends	Price Return (%)	Total Return (%)	BOP	
										Weight × Price Return (%)	BOP Weight × Total Return (%)
A	3,000	150,000	26.29	55.00	0.75	165,000	2,250	10.00	11.50	2.63	3.02
B	10,000	250,000	43.82	22.00	0.10	220,000	1,000	-12.00	-11.60	-5.26	-5.08
C	5,000	62,500	10.96	8.00	0.00	40,000	0	-36.00	-36.00	-3.95	-3.95
D	8,000	80,000	14.02	14.00	0.05	112,000	400	40.00	40.50	5.61	5.68
E	7,000	28,000	4.91	6.00	0.00	42,000	0	50.00	50.00	2.46	2.46
Total		570,500	100.00			579,000	3,650			1.49	2.13
Index Value		1,000				1,014.90	6.40	1.49	2.13		
Divisor = 570.50											
BOP = Beginning of period											
EOP = End of period											
Type of Index	BOP Value	Return (%)	EOP Value								
Price	1,000.00	1.49	1,014.90								
Return											
Total	1,000.00	2.13	1,021.30								
Return											

a total return of 11.50 percent, Security A contributes 5.66 percent to the total return of the price-weighted index, 2.30 percent to the total return of the equal-weighted index, and 3.02 percent to the total return of the market-capitalization-weighted index.

Float-Adjusted Market-Capitalization Weighting

In **float-adjusted market-capitalization weighting**, the weight on each constituent security is determined by adjusting its market capitalization for its **market float**. Typically, market float is the number of shares of the constituent security that are available to the investing public. For companies that are closely held, only a portion of the shares outstanding are available to the investing public (the rest are held by a small group of controlling investors). In addition to excluding shares held by controlling shareholders, most float-adjusted market-capitalization-weighted indexes also exclude shares held by other corporations and governments. Some providers of indexes that are designed to represent the investment opportunities of global investors further reduce the number of shares included in the index by excluding shares that are not available to foreigner investors. The index providers may refer to these indexes as “free-float-adjusted market-capitalization-weighted indexes.”

Float-adjusted market-capitalization-weighted indexes reflect the shares available for public trading by multiplying the market price per share by the number of shares available to the investing public (i.e., the float-adjusted market capitalization) rather than the total number of shares outstanding (total market capitalization). Currently, most market-capitalization-weighted indexes are float adjusted. Therefore, unless otherwise indicated, for the remainder of this reading, “market-capitalization” weighting refers to float-adjusted market-capitalization weighting.

The float-adjusted market-capitalization weight of security i is calculated as:

$$w_i^M = \frac{f_i Q_i P_i}{\sum_{j=1}^N f_j Q_j P_j} \quad (15)$$

where

f_i = fraction of shares outstanding in the market float

w_i = fraction of the portfolio that is allocated to security i or weight of security i

Q_i = number of shares outstanding of security i

P_i = share price of security i

N = number of securities in the index

Exhibit 5 illustrates the values, weights, and single-period returns following inception of a float-adjusted market-capitalization-weighted equity index using the same five securities as before. The low percentage of shares of Security D in the market float compared with the number of shares outstanding indicates that the security is closely held.

The primary advantage of market-capitalization weighting (including float adjusted) is that constituent securities are held in proportion to their value in the target market. The primary disadvantage is that constituent securities whose prices have risen the most (or fallen the most) have a greater (or lower) weight in the index (i.e., as a security's price rises relative to other securities in the index, its weight increases; and as its price decreases in value relative to other securities in the index, its weight decreases). This weighting method leads to overweighting stocks that have risen in price (and may be overvalued) and underweighting stocks that have declined in price (and may be undervalued). The effect of this weighting method is similar to a momentum investment strategy in that over time, the securities that have risen in price the most will have the largest weights in the index.

Exhibit 5: Example of Float-Adjusted Market-Capitalization-Weighted Equity Index

Stock	Shares Out-standing	Shares in Market	Shares in Index	BOP Price	BOP Float- Adjusted Market Cap	BOP Weight (%)	Dividends		Ending Float- Adjusted Market Cap	Total Dividends	Price Return (%)	Total Return (%)	BOP		EOP Weight (%)
							EOP Price	Dividends Per Share					Weight × Price Return (%)	Weight × Total Return (%)	
A	3,000	100	3,000	50.00	150,000	35.40	55.00	0.75	165,000	2,250	10.00	11.50	3.54	4.07	39.61
B	10,000	70	7,000	25.00	175,000	41.31	22.00	0.10	154,000	700	-12.00	-11.60	-4.96	-4.79	36.97
C	5,000	90	4,500	12.50	56,250	13.28	8.00	0.00	36,000	0	-36.00	-36.00	-4.78	-4.78	8.64
D	8,000	25	2,000	10.00	20,000	4.72	14.00	0.05	28,000	100	40.00	40.50	1.89	1.91	6.72
E	7,000	80	5,600	4.00	22,400	5.29	6.00	0.00	33,600	0	50.00	50.00	2.65	2.65	8.06
Total					423,650	100.00			416,600	3,050			-1.66	-0.94	100.00
Index Value					1,000				983.36	7.20	-1.66	-0.94			
Divisor = 423.65															
BOP = Beginning of period															
EOP = End of period															
Type of Index	Initial Value	Return (%)	Ending Value												
Price	1,000.00	-1.66	983.36												
Return															
Total	1,000.00	-0.94	990.56												
Return															

Fundamental Weighting

Fundamental weighting attempts to address the disadvantages of market-capitalization weighting by using measures of a company's size that are independent of its security price to determine the weight on each constituent security. These measures include book value, cash flow, revenues, earnings, dividends, and number of employees.

Some fundamental indexes use a single measure, such as total dividends, to weight the constituent securities, whereas others combine the weights from several measures to form a composite value that is used for weighting.

Letting F_i denote a given fundamental size measure of company i , the fundamental weight on security i is:

$$w_i^F = \frac{F_i}{\sum_{j=1}^N F_j} \quad (16)$$

Relative to a market-capitalization-weighted index, a fundamental index with weights based on such an item as earnings will result in greater weights on constituent securities with earnings yields (earnings divided by price) that are higher than the earnings yield of the overall market-weighted portfolio. Similarly, stocks with earnings yields less than the yield on the overall market-weighted portfolio will have lower weights. For example, suppose there are two stocks in an index. Stock A has a market capitalization of €200 million, Stock B has a market capitalization of €800 million, and their aggregate market capitalization is €1 billion (€1,000 million). Both companies have earnings of €20 million and aggregate earnings of €40 million. Thus, Stock A has an earnings yield of 10 percent (20/200) and Stock B has an earnings yield of 2.5 percent (20/800). The earnings weight of Stock A is 50 percent (20/40), which is higher than its market-capitalization weight of 20 percent (200/1,000). The earnings weight of Stock B is 50 percent (20/40), which is less than its market-capitalization weight of 80 percent (800/1,000). Relative to the market-cap-weighted index, the earnings-weighted index over-weights the high-yield Stock A and under-weights the low-yield Stock B.

The most important property of fundamental weighting is that it leads to indexes that have a “value” tilt. That is, a fundamentally weighted index has ratios of book value, earnings, dividends, etc. to market value that are higher than its market-capitalization-weighted counterpart. Also, in contrast to the momentum “effect” of market-capitalization-weighted indexes, fundamentally weighted indexes generally will have a contrarian “effect” in that the portfolio weights will shift away from securities that have increased in relative value and toward securities that have fallen in relative value whenever the portfolio is rebalanced.

INDEX MANAGEMENT: REBALANCING AND RECONSTITUTION

4



describe rebalancing and reconstitution of an index

So far, we have discussed index construction. Index management entails the two remaining questions:

- When should the index be rebalanced?
- When should the security selection and weighting decisions be re-examined?

Rebalancing

Rebalancing refers to adjusting the weights of the constituent securities in the index. To maintain the weight of each security consistent with the index's weighting method, the index provider rebalances the index by adjusting the weights of the constituent securities on a regularly scheduled basis (rebalancing dates)—usually quarterly. Rebalancing is necessary because the weights of the constituent securities change as their market prices change. Note, for example, that the weights of the securities in the equal-weighted index (Exhibit 3) at the end of the period are no longer equal (i.e., 20 percent):

Security A	19.93%
Security B	15.94
Security C	11.60
Security D	25.36
Security E	27.17

In rebalancing the index, the weights of Securities D and E (which had the highest returns) would be decreased and the weights of Securities A, B, and C (which had the lowest returns) would be increased. Thus, rebalancing creates turnover within an index.

Price-weighted indexes are not rebalanced because the weight of each constituent security is determined by its price. For market-capitalization-weighted indexes, rebalancing is less of a concern because the indexes largely rebalance themselves. In our market-capitalization index, for example, the weight of Security C automatically declined from 10.96 percent to 6.91 percent, reflecting the 36 percent decline in its market price. Market-capitalization weights are only adjusted to reflect mergers, acquisitions, liquidations, and other corporate actions between rebalancing dates.

Reconstitution

Reconstitution is the process of changing the constituent securities in an index. It is similar to a portfolio manager deciding to change the securities in his or her portfolio. Reconstitution is part of the rebalancing cycle. The reconstitution date is the date on which index providers review the constituent securities, re-apply the initial criteria for inclusion in the index, and select which securities to retain, remove, or add. Constituent securities that no longer meet the criteria are replaced with securities that do meet the criteria. Once the revised list of constituent securities is determined, the weighting method is re-applied. Indexes are reconstituted to reflect changes in the target market (bankruptcies, de-listings, mergers, acquisitions, etc.) and/or to reflect the judgment of the selection committee.

Reconstitution creates turnover in a number of different ways, particularly for market-capitalization-weighted indexes. When one security is removed and another is added, the index provider has to change the weights of the other securities in order to maintain the market-capitalization weighting of the index.

The frequency of reconstitution is a major issue for widely used indexes and their constituent securities. The Russell 2000 Index, for example, reconstitutes annually. It is used as a benchmark by numerous investment funds, and each year, prior to the index's reconstitution, the managers of these funds buy stocks they think will be added to the index—driving those stocks' prices up—and sell stocks they think will be deleted from the index—driving those stocks' prices down. Exhibit 6 illustrates a historical example of the potential impact of these decisions. Beginning in late April 2009, some managers began acquiring and bidding up the price of Uranium Energy Corporation (UEC) because they believed that it would be included in the reconstituted Russell 2000 Index. On 12 June, Russell listed UEC as a preliminary addition to

the Russell 2000 Index and the Russell 3000 Index.² By that time, the stock value had increased by more than 300 percent. Investors continued to bid up the stock price in the weeks following the announcement, and the stock closed on the reconstitution date of 30 June at USD2.90, up nearly 400 percent for the quarter.

Exhibit 6: Three-Month Performance of Uranium Energy Corporation and NASDAQ April through June 2009



Source: Yahoo! Finance and Capital IQ.

USES OF MARKET INDEXES

5

☐ describe uses of security market indexes

Indexes were initially created to give a sense of how a particular security market performed on a given day. With the development of modern financial theory, their uses in investment management have expanded significantly. Some of the major uses of indexes include:

- gauges of market sentiment;
- proxies for measuring and modeling returns, systematic risk, and risk-adjusted performance;
- proxies for asset classes in asset allocation models;
- benchmarks for actively managed portfolios; and
- model portfolios for such investment products as index funds and exchange-traded funds (ETFs).

² According to the press release, final membership in the index would be published after market close on Friday, 26 June.

Investors using security market indexes must be familiar with how various indexes are constructed in order to select the index or indexes most appropriate for their needs.

Gauges of Market Sentiment

The original purpose of stock market indexes was to provide a gauge of investor confidence or market sentiment. As indicators of the collective opinion of market participants, indexes reflect investor attitudes and behavior. The Dow Jones Industrial Average has a long history, is frequently quoted in the media, and remains a popular gauge of market sentiment. It may not accurately reflect the overall attitude of investors or the “market,” however, because the index consists of only 30 of the thousands of US stocks traded each day.

Proxies for Measuring and Modeling Returns, Systematic Risk, and Risk-Adjusted Performance

The capital asset pricing model (CAPM) defines beta as the systematic risk of a security with respect to the entire market. The market portfolio in the CAPM consists of all risky securities. To represent the performance of the market portfolio, investors use a broad index. For example, the Tokyo Price Index (TOPIX) and the S&P 500 often serve as proxies for the market portfolio in Japan and the United States, respectively, and are used for measuring and modeling systematic risk and market returns.

Security market indexes also serve as market proxies when measuring risk-adjusted performance. The beta of an actively managed portfolio allows investors to form a passive alternative with the same level of systematic risk. For example, if the beta of an actively managed portfolio of global stocks is 0.95 with respect to the MSCI World Index, investors can create a passive portfolio with the same systematic risk by investing 95 percent of their portfolio in a MSCI World Index fund and holding the remaining 5 percent in cash. Alpha, the difference between the return of the actively managed portfolio and the return of the passive portfolio, is a measure of risk-adjusted return or investment performance. Alpha can be the result of manager skill (or lack thereof), transaction costs, and fees.

Proxies for Asset Classes in Asset Allocation Models

Because indexes exhibit the risk and return profiles of select groups of securities, they play a critical role as proxies for asset classes in asset allocation models. They provide the historical data used to model the risks and returns of different asset classes.

Benchmarks for Actively Managed Portfolios

Investors often use indexes as benchmarks to evaluate the performance of active portfolio managers. The index selected as the benchmark should reflect the investment strategy used by the manager. For example, an active manager investing in global small-capitalization stocks should be evaluated using a benchmark index, such as the FTSE Global Small Cap Index, which includes approximately 4,400 liquid small-capitalization stocks across 47 countries as of August 2018.

The choice of an index to use as a benchmark is important because an inappropriate index could lead to incorrect conclusions regarding an active manager’s investment performance. Suppose that the small-cap manager underperformed the small-cap index but outperformed a broad equity market index. If investors use the broad market

index as a benchmark, they might conclude that the small-cap manager is earning his or her fees and should be retained or given additional assets to invest. Using the small-cap index as a benchmark might lead to a very different conclusion.

Model Portfolios for Investment Products

Indexes also serve as the basis for the development of new investment products. Using indexes as benchmarks for actively managed portfolios has led some investors to conclude that they should invest in the benchmarks instead. Based on the CAPM's conclusion that investors should hold the market portfolio, broad market index funds have been developed to function as proxies for the market portfolio.

Investment management firms initially developed and managed index portfolios for institutional investors. Eventually, mutual fund companies introduced index funds for individual investors. Subsequently, investment management firms introduced exchange-traded funds, which are managed the same way as index mutual funds but trade like stocks.

The first ETFs were based on existing indexes. As the popularity of ETFs increased, index providers created new indexes for the specific purpose of forming ETFs, leading to the creation of numerous narrowly defined indexes with corresponding ETFs. The VanEck Vectors Vietnam ETF, for example, allows investors to invest in the equity market of Vietnam.

The choice of indexes to meet the needs of investors is extensive. Index providers are constantly looking for opportunities to develop indexes to meet the needs of investors.

EQUITY INDEXES

6

- ☐ describe types of equity indexes
- ☐ compare types of security market indexes

A wide variety of equity indexes exist, including broad market, multi-market, sector, and style indexes.

Broad Market Indexes

A broad equity market index, as its name suggests, represents an entire given equity market and typically includes securities representing more than 90 percent of the selected market. For example, the Shanghai Stock Exchange Composite Index (SSE) is a market-capitalization-weighted index of all shares that trade on the Shanghai Stock Exchange. In the United States, the Wilshire 5000 Total Market Index is a market-capitalization-weighted index that includes all US equities with readily available prices and is designed to represent the entire US equity market.³ The Russell 3000, consisting of the largest 3,000 stocks by market capitalization, represents approximately 98 percent of the US equity market.

³ Despite its name, the Wilshire 5000 has no constraint on the number of securities that can be included. It included approximately 5,000 securities at inception.

Multi-Market Indexes

Multi-market indexes usually comprise indexes from different countries and regions and are designed to represent multiple security markets. Multi-market indexes may represent multiple national markets, geographic regions, economic development groups, and, in some cases, the entire world. World indexes are of importance to investors who take a global approach to equity investing without any particular bias toward a particular country or region. A number of index providers publish families of multi-market equity indexes.

MSCI offers a number of multi-market indexes. As shown in Exhibit 7, MSCI classifies countries and regions along two dimensions: level of economic development and geographic region. Developmental groups, which MSCI refers to as market classifications, include developed markets, emerging markets, and frontier markets. The geographic regions are largely divided by longitudinal lines of the globe: the Americas, Europe with Africa, and Asia with the Pacific. MSCI provides country- and region-specific indexes for each of the developed and emerging markets within its multi-market indexes. MSCI periodically reviews the classifications of markets in its indexes for movement from frontier markets to emerging markets and from emerging markets to developed markets and reconstitutes the indexes accordingly.

Exhibit 7: MSCI Global Investable Market Indexes (as of October 2018)

Developed Markets				
Americas		Europe and Middle East		Pacific
Canada, United States		Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Israel, Italy, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom		Australia, Hong Kong SAR, Japan, New Zealand, Singapore
Emerging Markets				
Americas		Europe, Middle East, Africa		Asia
Brazil, Chile, Colombia, Mexico, Peru		Czech Republic, Egypt, Greece, Hungary, Poland, Qatar, Russia, South Africa, Turkey, United Arab Emirates		Chinese mainland, India, Indonesia, South Korea, Malaysia, Pakistan, Philippines, Taiwan region, Thailand
Frontier Markets				
Americas	Europe & CIS	Africa	Middle East	Asia
Argentina	Croatia, Estonia, Lithuania, Kazakhstan, Romania, Serbia, Slovenia	Kenya, Mauritius, Morocco, Nigeria, Tunisia, WAEMU ¹	Bahrain, Jordan, Kuwait, Lebanon, Oman	Bangladesh, Sri Lanka, Vietnam
MSCI Standalone Market Indexes ²				
Europe, Middle East, and Africa	Americas	Europe and CIS	Africa	Middle East
Saudi Arabia	Jamaica, Panama, ³ Trinidad & Tobago	Bosnia Herzegovina, Bulgaria, Ukraine	Botswana, Ghana, Zimbabwe	Palestine

¹ The West African Economic and Monetary Union (WAEMU) consists of the following countries: Benin, Burkina Faso, Ivory Coast, Guinea-Bissau, Mali, Niger, Senegal, and Togo. Currently the MSCI WAEMU Indexes include securities classified in Senegal, Ivory Coast, and Burkina Faso.

² The MSCI Standalone Market Indexes are not included in the MSCI Emerging Markets Index or MSCI

Frontier Markets Index. However, these indexes use either the Emerging Markets or the Frontier Markets methodological criteria concerning size and liquidity.

³ *MSCI Panama Index has been launched as a Standalone Market Index.*

Source: adapted from MSCI (<https://www.msci.com/en/market-cap-weighted-indexes>), October 2018.

Fundamental Weighting in Multi-Market Indexes

Some index providers weight the securities within each country/region by market capitalization and then weight each country/region in the overall index in proportion to its relative GDP, effectively creating fundamental weighting in multi-market indexes. GDP-weighted indexes were some of the first fundamentally weighted indexes created. Introduced in 1987 by MSCI to address the 60 percent weight of Japanese equities in the market-capitalization-weighted MSCI EAFE Index at the time, GDP-weighted indexes reduced the allocation to Japanese equities by half.⁴

Sector Indexes

Sector indexes represent and track different economic sectors—such as consumer goods, energy, finance, health care, and technology—on either a national, regional, or global basis. Because different sectors of the economy behave differently over the course of the business cycle, some investors may seek to overweight or underweight their exposure to particular sectors.

Sector indexes are organized as families; each index within the family represents an economic sector. Typically, the aggregation of a sector index family is equivalent to a broad market index. Economic sector classification can be applied on a global, regional, or country-specific basis, but no universally agreed upon sector classification method exists.

Sector indexes play an important role in performance analysis because they provide a means to determine whether a portfolio manager is more successful at stock selection or sector allocation. Sector indexes also serve as model portfolios for sector-specific ETFs and other investment products.

Style Indexes

Style indexes represent groups of securities classified according to market capitalization, value, growth, or a combination of these characteristics. They are intended to reflect the investing styles of certain investors, such as the growth investor, value investor, and small-cap investor.

Market Capitalization

Market-capitalization indexes represent securities categorized according to the major capitalization categories: large cap, midcap, and small cap. With no universal definition of these categories, the indexes differ on the distinctions between large cap and midcap and between midcap and small cap, as well as the minimum market-capitalization size required to be included in a small-cap index. Classification into categories can be based on absolute market capitalization (e.g., below €100 million) or relative market capitalization (e.g., the smallest 2,500 stocks).

⁴ Steven A. Schoenfeld, *Active Index Investing* (Hoboken, NJ: John Wiley & Sons, 2004):220.

Value/Growth Classification

Some indexes represent categories of stocks based on their classifications as either value or growth stocks. Different index providers use different factors and valuation ratios (low price-to-book ratios, low price-to-earnings ratios, high dividend yields, etc.) to distinguish between value and growth equities.

Market Capitalization and Value/Growth Classification

Combining the three market-capitalization groups with value and growth classifications results in six basic style index categories:

- Large-Cap Value
- Mid-Cap Value
- Small-Cap Value
- Large-Cap Growth
- Mid-Cap Growth
- Small-Cap Growth

Because indexes use different size and valuation classifications, the constituents of indexes designed to represent a given style, such as small-cap value, may differ—sometimes substantially.

Because valuation ratios and market capitalizations change over time, stocks frequently migrate from one style index category to another on reconstitution dates. As a result, style indexes generally have much higher turnover than do broad market indexes.

7**FIXED-INCOME INDEXES**

- ☐ describe types of fixed-income indexes
- ☐ compare types of security market indexes

A wide variety of fixed-income indexes exists, but the nature of the fixed-income markets and fixed-income securities leads to some very important challenges to fixed-income index construction and replication. These challenges are the number of securities in the fixed-income universe, the availability of pricing data, and the liquidity of the securities.

Construction

The fixed-income universe includes securities issued by governments, government agencies, and corporations. Each of these entities may issue a variety of fixed-income securities with different characteristics. As a result, the number of fixed-income securities is many times larger than the number of equity securities. To represent a specific fixed-income market or segment, indexes may include thousands of different securities. Over time, these fixed-income securities mature, and issuers offer new securities to meet their financing needs, leading to turnover in fixed-income indexes.

Another challenge in index construction is that fixed-income markets are predominantly dealer markets. This means that firms (dealers) are assigned to specific securities and are responsible for creating liquid markets for those securities by purchasing and selling them from their inventory. In addition, many securities do not trade frequently and, as a result, are relatively illiquid. As a result, index providers

must contact dealers to obtain current prices on constituent securities to update the index or they must estimate the prices of constituent securities using the prices of traded fixed-income securities with similar characteristics.

These challenges can result in indexes with dissimilar numbers of bonds representing the same markets. The large number of fixed-income securities—combined with the lack of liquidity of some securities—has made it more costly and difficult, compared with equity indexes, for investors to replicate fixed-income indexes and duplicate their performance.

Types of Fixed-Income Indexes

The wide variety of fixed-income securities, ranging from zero-coupon bonds to bonds with embedded options (i.e., callable or putable bonds), results in a number of different types of fixed-income indexes. Similar to equities, fixed-income securities can be categorized according to the issuer's economic sector, the issuer's geographic region, or the economic development of the issuer's geographic region. Fixed-income securities can also be classified along the following dimensions:

- type of issuer (government, government agency, corporation);
- type of financing (general obligation, collateralized);
- currency of payments;
- maturity;
- credit quality (investment grade, high yield, credit agency ratings); and
- absence or presence of inflation protection.

Fixed-income indexes are based on these various dimensions and can be categorized as follows:

- aggregate or broad market indexes;
- market sector indexes;
- style indexes;
- economic sector indexes; and
- specialized indexes such as high-yield, inflation-linked, and emerging market indexes.

The first fixed-income index created, the Bloomberg Barclays US Aggregate Bond Index (formerly the Barclays Capital Aggregate Bond Index), is an example of a single-country aggregate index. Designed to represent the broad market of US fixed-income securities, it comprises approximately 8,000 securities, including US Treasury, government-related, corporate, mortgage-backed, asset-backed, and commercial mortgage-backed securities.

Aggregate indexes can be subdivided by market sector (government, government agency, collateralized, corporate); style (maturity, credit quality); economic sector, or some other characteristic to create more narrowly defined indexes. A common distinction reflected in indexes is between investment grade (e.g., those with a Standard & Poor's credit rating of BBB– or better) and high-yield securities. Investment-grade indexes are typically further subdivided by maturity (i.e., short, intermediate, or long) and by credit rating (e.g., AAA, BBB, etc.).⁵ The wide variety of fixed-income indexes reflects the partitioning of fixed-income securities on the basis of a variety of dimensions.

Exhibit 8 illustrates how the major types of fixed-income indexes can be organized on the basis of various dimensions.

⁵ Credit ratings are discussed in depth in the Level I CFA Program reading "Fundamentals of Credit Analysis."

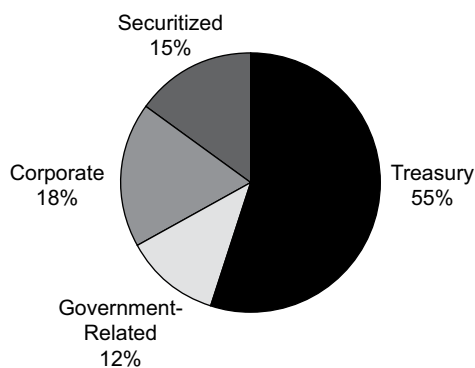
Exhibit 8: Dimensions of Fixed-Income Indexes

Market	Global			
	Regional			
	Country or currency zone			
Type	Corporate	Collateralized <i>Securitized</i> <i>Mortgage-backed</i>	Government agency	Government
Maturity	For example, 1–3, 3–5, 5–7, 7–10, 10+ years; short-term, medium-term, or long-term			
Credit quality	For example, AAA, AA, A, BBB, etc.; Aaa, Aa, A, Baa, etc.; investment grade, high yield			

All aggregate indexes include a variety of market sectors and credit ratings. The breakdown of the Bloomberg Barclays Global Aggregate Bond Index by market sectors and by credit rating is shown in Exhibit 9 and Exhibit 10, respectively.

Exhibit 9: Market Sector Breakdown of the Bloomberg Barclays Global Aggregate Bond Index

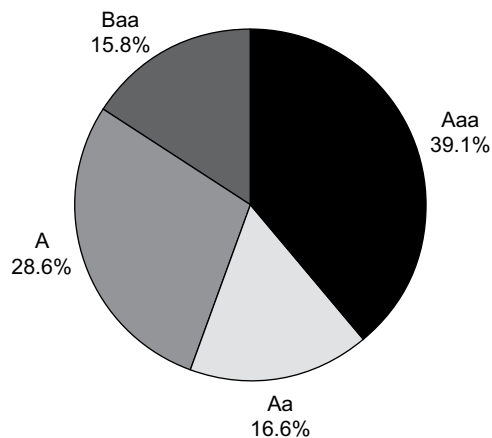
Sector Breakdown as of June 30, 2016



Source: Bloomberg Barclays Indices, Global Aggregate Index Factsheet, August 24, 2016. <https://data.bloomberglp.com/indices/sites/2/2016/08/Factsheet-Global-Aggregate.pdf>.

Exhibit 10: Credit Breakdown of the Bloomberg Barclays Global Aggregate Bond Index

Quality Breakdown as of June 30, 2016



Source: Bloomberg Barclays Indices, Global Aggregate Index Factsheet, August 24, 2016. <https://data.bloomberglp.com/indices/sites/2/2016/08/Factsheet-Global-Aggregate.pdf>.

INDEXES FOR ALTERNATIVE INVESTMENTS

8

- ☐ describe indexes representing alternative investments
- ☐ compare types of security market indexes

Many investors seek to lower the risk or enhance the performance of their portfolios by investing in assets classes other than equities and fixed income. Interest in alternative assets and investment strategies has led to the creation of indexes designed to represent broad classes of alternative investments. Three of the most widely followed alternative investment classes are commodities, real estate, and hedge funds.

Commodity Indexes

Commodity indexes consist of futures contracts on one or more commodities, such as agricultural products (rice, wheat, sugar), livestock (cattle, hogs), precious and common metals (gold, silver, copper), and energy commodities (crude oil, natural gas).

Although some commodity indexes may include the same commodities, the returns of these indexes may differ because each index may use a different weighting method. Because commodity indexes do not have an obvious weighting mechanism, such as market capitalization, commodity index providers create their own weighting methods. Some indexes, such as the Thomson Reuters/Core Commodity CRB Index (TR/CC CRB Index), formerly known as the Commodity Research Bureau (CRB) Index, contain a fixed number of commodities that are weighted equally. The S&P

GSCI uses a combination of liquidity measures and world production values in its weighting scheme and allocates more weight to commodities that have risen in price. Other indexes have fixed weights that are determined by a committee.

The different weighting methods can also lead to large differences in exposure to specific commodities. The S&P GSCI in 2018, for example, weights the energy-sector approximately 50% higher and the agriculture sector 40% lower than the CRB Index. These differences result in indexes with very different risk and return profiles. Unlike commodity indexes, broad equity and fixed-income indexes that target the same markets share similar risk and return profiles.

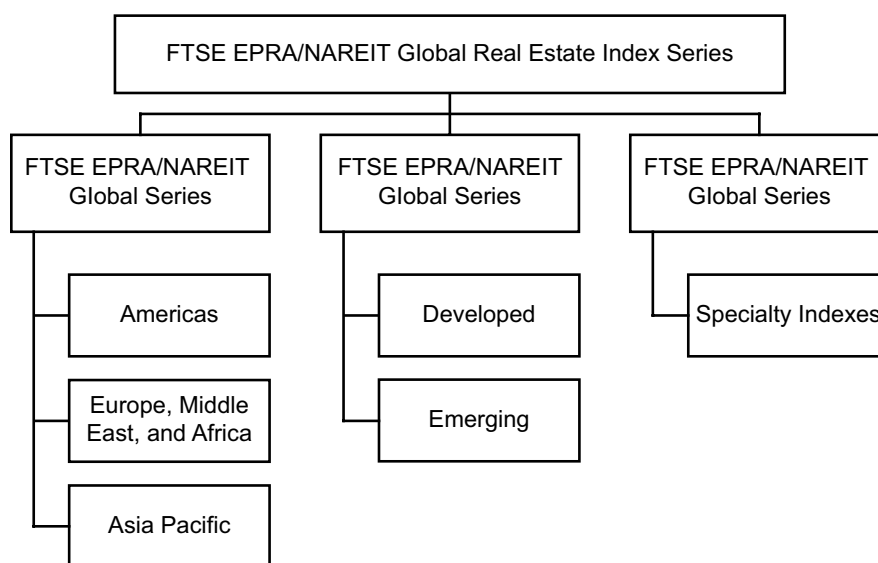
The performance of commodity indexes can also be quite different from their underlying commodities because the indexes consist of futures contracts on the commodities rather than the actual commodities. Index returns are affected by factors other than changes in the prices of the underlying commodities because futures contracts must be continually “rolled over” (i.e., replacing a contract nearing expiration with a new contract). Commodity index returns reflect the risk-free interest rate, the changes in future prices, and the roll yield. Therefore, a commodity index return can be quite different from the return based on changes in the prices of the underlying commodities.

Real Estate Investment Trust Indexes

Real estate indexes represent not only the market for real estate securities but also the market for real estate—a highly illiquid market and asset class with infrequent transactions and pricing information. Real estate indexes can be categorized as appraisal indexes, repeat sales indexes, and real estate investment trust (REIT) indexes.

REIT indexes consist of shares of publicly traded REITs. REITs are public or private corporations organized specifically to invest in real estate, either through ownership of properties or investment in mortgages. Shares of public REITs are traded on the world's various stock exchanges and are a popular choice for investing in commercial real estate properties. Because REIT indexes are based on publicly traded REITs with continuous market pricing, the value of REIT indexes is calculated continuously.

The FTSE EPRA/NAREIT global family of REIT indexes shown in Exhibit 11 seeks to represent trends in real estate stocks worldwide and includes representation from the European Public Real Estate Association (EPRA) and the National Association of Real Estate Investment Trusts (NAREIT).

Exhibit 11: The FTSE EPRA/NAREIT Global REIT Index Family

Source: FTSE International, “FTSE EPRA/NAREIT Global & Global Ex US Indices” Factsheet 2009). “FTSE” is a trade mark of the London Stock Exchange Plc, “NAREIT” is a trade mark of the National Association of Real Estate Investment Trusts (“NAREIT”) and “EPRA” is a trade mark of the European Public Real Estate Association (“EPRA”) and all are used by FTSE International Limited (“FTSE”) under license.

Hedge Fund Indexes

Hedge fund indexes reflect the returns on hedge funds. **Hedge funds** are private investment vehicles that typically use leverage and long and short investment strategies.

A number of research organizations maintain databases of hedge fund returns and summarize these returns into indexes. These database indexes are designed to represent the performance of the hedge funds on a very broad global level (hedge funds in general) or the strategy level. Most of these indexes are equal weighted and represent the performance of the hedge funds within a particular database.

Most research organizations rely on the voluntary cooperation of hedge funds to compile performance data. As unregulated entities, however, hedge funds are not required to report their performance to any party other than their investors. Therefore, each hedge fund decides to which database(s) it will report its performance. As a result, rather than index providers determining the constituents, the constituents determine the index.

Frequently, a hedge fund reports its performance to only one database. The result is little overlap of funds covered by the different indexes. With little overlap between their constituents, different global hedge fund indexes may reflect very different performance for the hedge fund industry over the same period of time.

Another consequence of the voluntary performance reporting is the potential for survivorship bias and, therefore, inaccurate performance representation. This means that hedge funds with poor performance may be less likely to report their performance to the database or may stop reporting to the database, so their returns may be excluded when measuring the return of the index. As a result, the index may not accurately reflect actual hedge fund performance so much as the performance of hedge funds that are performing well.

REPRESENTATIVE INDEXES WORLDWIDE

As indicated in this reading, the choice of indexes to meet the needs of investors is extensive. Investors using security market indexes must be careful in their selection of the index or indexes most appropriate for their needs. The following table illustrates the variety of indexes reflecting different asset classes, markets, and weighting methods.

Index	Representing	Number of Securities	Weighting Method	Comments
Dow Jones Industrial Average	US blue chip companies	30	Price	The oldest and most widely known US equity index. <i>Wall Street Journal</i> editors choose 30 stocks from among large, mature blue-chip companies.
Nikkei Stock Average	Japanese blue chip companies	225	Modified price	Known as the Nikkei 225 and originally formulated by Dow Jones & Company. Because of extreme variation in price levels of component securities, some high-priced shares are weighted as a fraction of share price.
TOPIX	All companies listed on the Tokyo Stock Exchange First Section	Varies	Float-adjusted market cap	Represents about 93 percent of the market value of all Japanese equities. Contains a large number of very small, illiquid stocks, making exact replication difficult.
MSCI All Country World Index	Stocks of 23 developed and 24 emerging markets	Varies	Free-float-adjusted market cap	Composed of companies representative of the market structure of developed and emerging market countries in the Americas, Europe/Middle East, and Asia/Pacific regions. Price return and total return versions available in both USD and local currencies.
S&P Developed Ex-US BMI Energy Sector Index	Energy sector of developed global markets outside the United States	Varies	Float-adjusted market cap	Serves as a model portfolio for the SPDR® S&P Energy Sector Exchange-Traded Fund (ETF).
Bloomberg Barclays Global Aggregate Bond Index	Investment-grade bonds in the North American, European, and Asian markets	Varies	Market cap	Formerly known as Lehman Brothers Global Aggregate Bond Index.
Markit iBoxx Euro High-Yield Bond Indexes	Sub-investment-grade euro-denominated corporate bonds	Varies	Market cap and variations	Rebalanced monthly. Represents tradable part of market. Price and total return versions available with such analytical values as yield, duration, modified duration, and convexity. Provides platform for research and structured products.
FTSE EPRA/NAREIT Global Real Estate Index	Real estate securities in the North American, European, and Asian markets	Varies	Float-adjusted market cap	The stocks of REITs that constitute the index trade on public stock exchanges and may be constituents of equity market indexes.

Index	Representing	Number of Securities	Weighting Method	Comments
HFRX Global Hedge Fund Index	Overall composition of the HFR database	Varies	Asset weighted	Comprises all eligible hedge fund strategies. Examples include convertible arbitrage, distressed securities, market neutral, event driven, macro, and relative value arbitrage. Constituent strategies are asset weighted on the basis of asset distribution within the hedge fund industry.
HFRX Equal Weighted Strategies EUR Index	Overall composition of the HFR database	Varies	Equal weighted	Denominated in euros and is constructed from the same strategies as the HFRX Global Hedge Fund Index.
Morningstar Style Indexes	US stocks classified by market cap and value/growth orientation	Varies	Float-adjusted market cap	The nine indexes defined by combinations of market cap (large, mid, and small) and value/growth orientation (value, core, growth) have mutually exclusive constituents and are exhaustive with respect to the Morningstar US Market Index. Each is a model portfolio for one of the iShares Morningstar ETFs.

SUMMARY

This reading explains and illustrates the construction, management, and uses of security market indexes. It also discusses various types of indexes. Security market indexes are invaluable tools for investors, who can select from among thousands of indexes representing a variety of security markets, market segments, and asset classes. These indexes range from those representing the global market for major asset classes to those representing alternative investments in specific geographic markets. To benefit from the use of security market indexes, investors must understand their construction and determine whether the selected index is appropriate for their purposes. Frequently, an index that is well suited for one purpose may not be well suited for other purposes. Users of indexes must be familiar with how various indexes are constructed in order to select the index or indexes most appropriate for their needs.

Among the key points made in this reading are the following:

- Security market indexes are intended to measure the values of different target markets (security markets, market segments, or asset classes).
- The constituent securities selected for inclusion in the security market index are intended to represent the target market.
- A price return index reflects only the prices of the constituent securities.
- A total return index reflects not only the prices of the constituent securities but also the reinvestment of all income received since the inception of the index.

- Methods used to weight the constituents of an index range from the very simple, such as price and equal weightings, to the more complex, such as market-capitalization and fundamental weightings.
- Choices in index construction—in particular, the choice of weighting method—affect index valuation and returns.
- Index management includes 1) periodic rebalancing to ensure that the index maintains appropriate weightings and 2) reconstitution to ensure the index represents the desired target market.
- Rebalancing and reconstitution create turnover in an index. Reconstitution can dramatically affect prices of current and prospective constituents.
- Indexes serve a variety of purposes. They gauge market sentiment and serve as benchmarks for actively managed portfolios. They act as proxies for measuring systematic risk and risk-adjusted performance. They also serve as proxies for asset classes in asset allocation models and as model portfolios for investment products.
- Investors can choose from security market indexes representing various asset classes, including equity, fixed-income, commodity, real estate, and hedge fund indexes.
- Within most asset classes, index providers offer a wide variety of indexes, ranging from broad market indexes to highly specialized indexes based on the issuer's geographic region, economic development group, or economic sector or other factors.
- Proper use of security market indexes depends on understanding their construction and management.

PRACTICE PROBLEMS

1. A security market index represents the:
 - A. risk of a security market.
 - B. security market as a whole.
 - C. security market, market segment, or asset class.
2. One month after inception, the price return version and total return version of a single index (consisting of identical securities and weights) will be equal if:
 - A. market prices have not changed.
 - B. capital gains are offset by capital losses.
 - C. the securities do not pay dividends or interest.
3. The values of a price return index and a total return index consisting of identical equal-weighted dividend-paying equities will be equal:
 - A. only at inception.
 - B. at inception and on rebalancing dates.
 - C. at inception and on reconstitution dates.
4. Security market indexes are:
 - A. constructed and managed like a portfolio of securities.
 - B. simple interchangeable tools for measuring the returns of different asset classes.
 - C. valued on a regular basis using the actual market prices of the constituent securities.
5. When creating a security market index, an index provider must first determine the:
 - A. target market.
 - B. appropriate weighting method.
 - C. number of constituent securities.
6. An analyst gathers the following information for an equal-weighted index comprised of assets Able, Baker, and Charlie:

Security	Beginning of Period Price (€)	End of Period Price (€)	Total Dividends (€)
Able	10.00	12.00	0.75
Baker	20.00	19.00	1.00
Charlie	30.00	30.00	2.00

The price return of the index is:

- A. 1.7%.
- B. 5.0%.
- C. 11.4%.

7. An analyst gathers the following information for an equal-weighted index comprised of assets Able, Baker, and Charlie:

Security	Beginning of Period Price (€)	End of Period Price (€)	Total Dividends (€)
Able	10.00	12.00	0.75
Baker	20.00	19.00	1.00
Charlie	30.00	30.00	2.00

The total return of the index is:

- A. 5.0%.
- B. 7.9%.
- C. 11.4%.

8. An analyst gathers the following information for a price-weighted index comprised of securities ABC, DEF, and GHI:

Security	Beginning of Period Price (£)	End of Period Price (£)	Total Dividends (£)
ABC	25.00	27.00	1.00
DEF	35.00	25.00	1.50
GHI	15.00	16.00	1.00

The price return of the index is:

- A. -4.6%.
- B. -9.3%.
- C. -13.9%.

9. An analyst gathers the following information for a market-capitalization-weighted index comprised of securities MNO, QRS, and XYZ:

Security	Beginning of Period Price (¥)	End of Period Price (¥)	Dividends per Share (¥)	Shares Outstanding
MNO	2,500	2,700	100	5,000
QRS	3,500	2,500	150	7,500
XYZ	1,500	1,600	100	10,000

The price return of the index is:

- A. -9.33%.
- B. -10.23%.

C. -13.90%.

10. An analyst gathers the following information for a market-capitalization-weighted index comprised of securities MNO, QRS, and XYZ:

Security	Beginning of Period Price (¥)	End of Period Price (¥)	Dividends Per Share (¥)	Shares Outstanding
MNO	2,500	2,700	100	5,000
QRS	3,500	2,500	150	7,500
XYZ	1,500	1,600	100	10,000

The total return of the index is:

- A. 1.04%.
- B. -5.35%.
- C. -10.23%.
11. When creating a security market index, the target market:
- A. determines the investment universe.
- B. is usually a broadly defined asset class.
- C. determines the number of securities to be included in the index.

12. An analyst gathers the following data for a price-weighted index:

Security	Beginning of Period		End of Period	
	Price (€)	Shares Outstanding	Price (€)	Shares Outstanding
A	20.00	300	22.00	300
B	50.00	300	48.00	300
C	26.00	2,000	30.00	2,000

The price return of the index over the period is:

- A. 4.2%.
- B. 7.1%.
- C. 21.4%.

13. An analyst gathers the following data for a value-weighted index:

Security	Beginning of Period		End of Period	
	Price (£)	Shares Outstanding	Price (£)	Shares Outstanding
A	20.00	300	22.00	300
B	50.00	300	48.00	300
C	26.00	2,000	30.00	2,000

The return on the value-weighted index over the period is:

- A. 7.1%.
- B. 11.0%.
- C. 21.4%.

14. An analyst gathers the following data for an equally-weighted index:

Security	Beginning of Period		End of Period	
	Price (¥)	Shares Outstanding	Price (¥)	Shares Outstanding
A	20.00	300	22.00	300
B	50.00	300	48.00	300
C	26.00	2,000	30.00	2,000

The return on the index over the period is:

- A. 4.2%.
- B. 6.8%.
- C. 7.1%.

15. Which of the following index weighting methods requires an adjustment to the divisor after a stock split?

- A. Price weighting.
- B. Fundamental weighting.
- C. Market-capitalization weighting.

16. If the price return of an equal-weighted index exceeds that of a market-capitalization-weighted index comprised of the same securities, the *most likely* explanation is:

- A. stock splits.
- B. dividend distributions.
- C. outperformance of small-market-capitalization stocks.

17. A float-adjusted market-capitalization-weighted index weights each of its constit-

uent securities by its price and:

- A. its trading volume.
- B. the number of its shares outstanding.
- C. the number of its shares available to the investing public.

18. Which of the following index weighting methods is most likely subject to a value tilt?

- A. Equal weighting.
- B. Fundamental weighting.
- C. Market-capitalization weighting.

19. Rebalancing an index is the process of periodically adjusting the constituent:

- A. securities' weights to optimize investment performance.
- B. securities to maintain consistency with the target market.
- C. securities' weights to maintain consistency with the index's weighting method.

20. Which of the following index weighting methods requires the most frequent rebalancing?

- A. Price weighting.
- B. Equal weighting.
- C. Market-capitalization weighting.

21. Reconstitution of a security market index reduces:

- A. portfolio turnover.
- B. the need for rebalancing.
- C. the likelihood that the index includes securities that are not representative of the target market.

22. Security market indexes are used as:

- A. measures of investment returns.
- B. proxies to measure unsystematic risk.
- C. proxies for specific asset classes in asset allocation models.

23. Uses of market indexes do not include serving as a:

- A. measure of systemic risk.
- B. basis for new investment products.
- C. benchmark for evaluating portfolio performance.

24. Which of the following statements regarding sector indexes is *most* accurate?

Sector indexes:

- A. track different economic sectors and cannot be aggregated to represent the equivalent of a broad market index.
- B. provide a means to determine whether an active investment manager is more successful at stock selection or sector allocation.
- C. apply a universally agreed upon sector classification system to identify the constituent securities of specific economic sectors, such as consumer goods, energy, finance, health care.

25. Which of the following is an example of a style index? An index based on:

- A. geography.
- B. economic sector.
- C. market capitalization.

26. Which of the following statements regarding fixed-income indexes is *most* accurate?

- A. Liquidity issues make it difficult for investors to easily replicate fixed-income indexes.
- B. Rebalancing and reconstitution are the only sources of turnover in fixed-income indexes.
- C. Fixed-income indexes representing the same target market hold similar numbers of bonds.

27. An aggregate fixed-income index:

- A. comprises corporate and asset-backed securities.
- B. represents the market of government-issued securities.
- C. can be subdivided by market or economic sector to create more narrowly defined indexes.

28. Fixed-income indexes are *least likely* constructed on the basis of:

- A. maturity.
- B. type of issuer.
- C. coupon frequency.

29. In comparison to equity indexes, the constituent securities of fixed-income indexes are:

- A. more liquid.
- B. easier to price.
- C. drawn from a larger investment universe.

30. Commodity index values are based on:

- A. futures contract prices.

- B. the market price of the specific commodity.
 - C. the average market price of a basket of similar commodities.
31. Which of the following statements is *most* accurate?
- A. Commodity indexes all share similar weighting methods.
 - B. Commodity indexes containing the same underlying commodities offer similar returns.
 - C. The performance of commodity indexes can be quite different from that of the underlying commodities.
32. Which of the following is *not* a real estate index category?
- A. Appraisal index.
 - B. Initial sales index.
 - C. Repeat sales index.
33. A unique feature of hedge fund indexes is that they:
- A. are frequently equal weighted.
 - B. are determined by the constituents of the index.
 - C. reflect the value of private rather than public investments.
34. The returns of hedge fund indexes are *most likely*:
- A. biased upward.
 - B. biased downward.
 - C. similar across different index providers.

SOLUTIONS

1. C is correct. A security market index represents the value of a given security market, market segment, or asset class.
2. C is correct. The difference between a price return index and a total return index consisting of identical securities and weights is the income generated over time by the underlying securities. If the securities in the index do not generate income, both indexes will be identical in value.
3. A is correct. At inception, the values of the price return and total return versions of an index are equal.
4. A is correct. Security market indexes are constructed and managed like a portfolio of securities.
5. A is correct. The first decision is identifying the target market that the index is intended to represent because the target market determines the investment universe and the securities available for inclusion in the index.
6. B is correct. The price return is the sum of the weighted returns of each security. The return of Able is 20 percent $[(12 - 10)/10]$; of Baker is -5 percent $[(19 - 20)/20]$; and of Charlie is 0 percent $[(30 - 30)/30]$. The price return index assigns a weight of 1/3 to each asset; therefore, the price return is $1/3 \times [20\% + (-5\%) + 0\%] = 5\%$.
7. C is correct. The total return of an index is calculated on the basis of the change in price of the underlying securities plus the sum of income received or the sum of the weighted total returns of each security. The total return of Able is 27.5 percent; of Baker is 0 percent; and of Charlie is 6.7 percent:

$$\text{Able: } (12 - 10 + 0.75)/10 = 27.5\%$$

$$\text{Baker: } (19 - 20 + 1)/20 = 0\%$$

$$\text{Charlie: } (30 - 30 + 2)/30 = 6.7\%$$

An equal-weighted index applies the same weight (1/3) to each security's return; therefore, the total return = $1/3 \times (27.5\% + 0\% + 6.7\%) = 11.4\%$.

8. B is correct. The price return of the price-weighted index is the percentage change in price of the index: $(68 - 75)/75 = -9.33\%$.

Security	Beginning of Period Price (£)	End of Period Price (£)
ABC	25.00	27.00
DEF	35.00	25.00
GHI	15.00	16.00
TOTAL	75.00	68.00

9. B is correct. The price return of the index is $(48,250,000 - 53,750,000)/53,750,000 = -10.23\%$.

Security	Beginning of Period Price (¥)	Shares Outstanding	Beginning of Period Value (¥)	End of Period Price (¥)	End of Period Value (¥)
MNO	2,500	5,000	12,500,000	2,700	13,500,000
QRS	3,500	7,500	26,250,000	2,500	18,750,000
XYZ	1,500	10,000	15,000,000	1,600	16,000,000
Total			53,750,000		48,250,000

10. B is correct. The total return of the market-capitalization-weighted index is calculated below:

Security	Beginning of Period Value (¥)	End of Period Value (¥)	Total Dividends (¥)	Total Return (%)
MNO	12,500,000	13,500,000	500,000	12.00
QRS	26,250,000	18,750,000	1,125,000	-24.29
XYZ	15,000,000	16,000,000	1,000,000	13.33
Total	53,750,000	48,250,000	2,625,000	-5.35

11. A is correct. The target market determines the investment universe and the securities available for inclusion in the index.
12. A is correct. The sum of prices at the beginning of the period is 96; the sum at the end of the period is 100. Regardless of the divisor, the price return is $100/96 - 1 = 0.042$ or 4.2 percent.
13. B is correct. It is the percentage change in the market value over the period:

Market value at beginning of period: $(20 \times 300) + (50 \times 300) + (26 \times 2,000) = 73,000$

Market value at end of period: $(22 \times 300) + (48 \times 300) + (30 \times 2,000) = 81,000$

Percentage change is $81,000/73,000 - 1 = 0.1096$ or 11.0 percent with rounding.

14. C is correct. With an equal-weighted index, the same amount is invested in each security. Assuming \$1,000 is invested in each of the three stocks, the index value is \$3,000 at the beginning of the period and the following number of shares is purchased for each stock:

Security A: 50 shares

Security B: 20 shares

Security C: 38.46 shares.

Using the prices at the beginning of the period for each security, the index value at the end of the period is \$3,213.8: $(\$22 \times 50) + (\$48 \times 20) + (\$30 \times 38.46)$. The price return is $\$3,213.8/\$3,000 - 1 = 7.1\%$.

15. A is correct. In the price weighting method, the divisor must be adjusted so the index value immediately after the split is the same as the index value immediately prior to the split.
16. C is correct. The main source of return differences arises from outperformance

of small-cap securities or underperformance of large-cap securities. In an equal-weighted index, securities that constitute the largest fraction of the market are underrepresented and securities that constitute only a small fraction of the market are overrepresented. Thus, higher equal-weighted index returns will occur if the smaller-cap equities outperform the larger-cap equities.

17. C is correct. “Float” is the number of shares available for public trading.
18. B is correct. Fundamental weighting leads to indexes that have a value tilt.
19. C is correct. Rebalancing refers to adjusting the weights of constituent securities in an index to maintain consistency with the index’s weighting method.
20. B is correct. Changing market prices will cause weights that were initially equal to become unequal, thus requiring rebalancing.
21. C is correct. Reconstitution is the process by which index providers review the constituent securities, re-apply the initial criteria for inclusion in the index, and select which securities to retain, remove, or add. Constituent securities that no longer meet the criteria are replaced with securities that do. Thus, reconstitution reduces the likelihood that the index includes securities that are not representative of the target market.
22. C is correct. Security market indexes play a critical role as proxies for asset classes in asset allocation models.
23. A is correct. Security market indexes are used as proxies for measuring market or systematic risk, not as measures of systemic risk.
24. B is correct. Sector indexes provide a means to determine whether a portfolio manager is more successful at stock selection or sector allocation.
25. C is correct. Style indexes represent groups of securities classified according to market capitalization, value, growth, or a combination of these characteristics.
26. A is correct. The large number of fixed-income securities—combined with the lack of liquidity of some securities—makes it costly and difficult for investors to replicate fixed-income indexes.
27. C is correct. An aggregate fixed-income index can be subdivided by market sector (government, government agency, collateralized, corporate), style (maturity, credit quality), economic sector, or some other characteristic to create more narrowly defined indexes.
28. C is correct. Coupon frequency is not a dimension on which fixed-income indexes are based.
29. C is correct. The fixed-income market has more issuers and securities than the equity market.
30. A is correct. Commodity indexes consist of futures contracts on one or more commodities.
31. C is correct. The performance of commodity indexes can be quite different from that of the underlying commodities because the indexes consist of futures contracts on the commodities rather than the actual commodities.
32. B is correct. It is not a real estate index category.

33. B is correct. Hedge funds are not required to report their performance to any party other than their investors. Therefore, each hedge fund decides to which database(s) it will report its performance. Thus, for a hedge fund index, constituents determine the index rather than index providers determining the constituents.
34. A is correct. Voluntary performance reporting may lead to survivorship bias, and poorer performing hedge funds will be less likely to report their performance.

